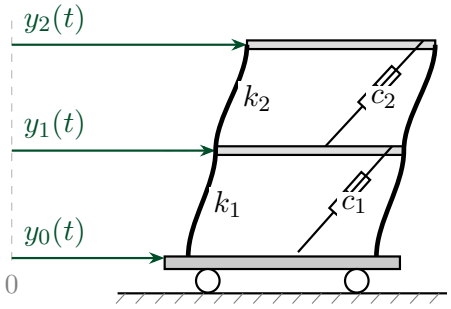
MA3521 — Building Models and Linear ODEs

Name:

1. **Two-Story Building Practice**

Displacements: $\mathbf{y}(t) = (y_1(t), y_2(t))$ (m), t in seconds.



$m_2 = 2\,\text{kg}$
$k_2 = 4\,\text{N/m}$
$c_2 = 2\,\text{N s/m}$

$m_1 = 3\,\text{kg}$
$k_1 = 6\,\text{N/m}$
$c_1 = 3\,\text{N s/m}$

$y_0(t) = 2\sin(t)$ m

The base motion $y_0(t)$ is prescribed. The floor displacements $y_1(t)$ and $y_2(t)$ are measured from rest.

(a) Fill in the matrices and forcing vector for $M\mathbf{y}'' + C\mathbf{y}' + K\mathbf{y} = \mathbf{b}(t)$. Leave $\mathbf{b}(t)$ in terms of $y_0(t)$ and $y_0'(t)$.

$$M = \begin{pmatrix} & \\ & \end{pmatrix} \quad C = \begin{pmatrix} & \\ & \end{pmatrix} \quad K = \begin{pmatrix} & \\ & \end{pmatrix} \quad \mathbf{b}(t) = \begin{pmatrix} \\ \end{pmatrix}$$

(b) With $\mathbf{z} = (y_1, y_2, y_1', y_2')$, fill in the matrix A for $\mathbf{z}' = A\mathbf{z} + \mathbf{g}(t)$.

$$A = \begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}$$

2. **Linear Initial Value Problem**

$$y' - \frac{4}{t}y = 2t^5 \qquad y(1) = 3 \qquad t > 0$$

(a) Find an integrating factor.

$$\mu(t) = \exp\left(\int \boxed{}\, dt\right) = \boxed{}$$

(b) Multiply by $\mu(t)$ and integrate.

$$(\mu y)' = \boxed{}$$

$$\mu(t)y(t) = \int \boxed{}\, dt + C = \boxed{} + C$$

(c) Write the solution.

$$\boxed{}$$